



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

FURTHER MATHEMATICS

PAPER 2

9187/2

NOVEMBER 2011 SESSION

3 hours

Additional materials:
Answer paper
Graph paper
List of Formulae

TIME 3 hours

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

There is no restriction on the number of questions which you may attempt.

If a numerical answer cannot be given exactly, and the accuracy required is not specified in the question, then in the case of an angle it should be given to the nearest degree, and in other cases it should be given correct to 2 significant figures.

If a numerical value for g is necessary, take $g = 9.81 \text{ ms}^{-2}$.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 120.

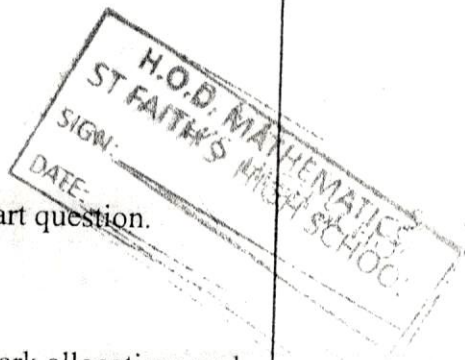
Within each section of the paper, questions are printed in the order of their mark allocations and candidates are advised, within each section, to attempt questions sequentially.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

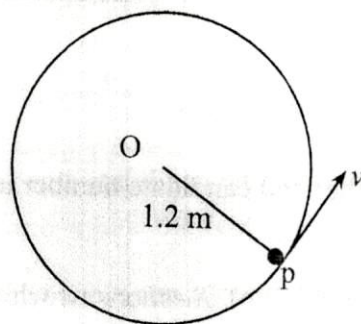
This question paper consists of 6 printed pages and 2 blank pages.

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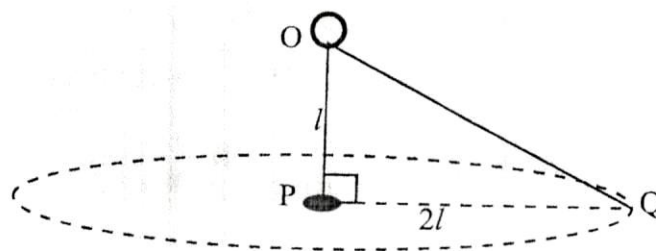
Section (a) : Mechanics

- 1 The engine of a 500 kg car is switched off at the instance it is about to go up a ramp inclined at $\sin^{-1}\left(\frac{2}{5}\right)$. If the car is travelling at 100 km/h at that instance and experiences a constant resistance of 50 N, calculate the distance it goes up the ramp. [4]
- 2 A particle of mass 0.2 kg is attached to an extensible string of natural length 2 m and modulus of elasticity 480 N. The other end is fixed to a point, C, on a smooth horizontal table. The particle is projected from C, with a velocity of 3 ms^{-1} . Find the maximum displacement of the particle from C. [4]
- 3 A ball bearing P of mass 0.2 kg is moving inside a smooth hollow sphere of radius 1.2 m in a vertical plane.



Find the least speed, v , of the particle at its lowest point so that it completes a circle in the sphere. [6]

H.O.D.
STAFF
DATE: / /



Two particles, P and Q, are connected by a light inextensible string passing through a smooth ring O. O is vertically above P and the string is taut. P and Q remain in a horizontal plane.

The masses of P and Q are M and m respectively. Q is made to move in a horizontal circle of radius $2l$ (see diagram above).

Find

- (a) the ratio $M : m$ [6]
- (b) the angular velocity of Q in terms of g and l . [3]

- 5 (a) A particle is executing simple harmonic motion in a straight line between the extreme points, R and S, and passes through the origin O.

If the distance $RS = 12$ cm and the maximum speed of the particle is 18 cm s^{-1} , find the period of the motion. [4]

- (b) A particle, P, describes simple harmonic motion centred at O with a period of 2 s and amplitude 2.5 cm.

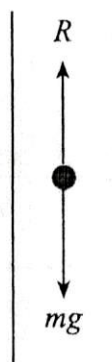
Point Q is 0.5 cm from O and point R is 0.8 cm from O.

Calculate the time interval for P to move directly from Q to R. [6]

- 6 Two spheres, X and Y, each of mass 5 kg lie at rest on a smooth horizontal plane. A third sphere, R, of the same radius as the other two but of mass 1 kg, moves with velocity, u , along the line of the centres of X and Y. The sphere R collides with P which in turn collides with Y.

- (a) If R is brought to rest by the first collision, show that the coefficient of restitution between R and Y is $\frac{1}{5}$. [3]
- (b) If the coefficient of restitution between X and Y is $\frac{1}{3}$, find the velocities of X and Y after the impact. [5]
- (c) Show that the total loss of kinetic energy due to the two collisions is $\frac{4}{9}u^2$. [4]

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A ball bearing of mass m falling through oil column experiences a resistive force proportional to its velocity. This means that the resistive force, $R = kv$, where k is the constant of proportionality (see diagram).

- (a) Show that the velocity of the ball bearing satisfies $\frac{dv}{dt} = g - \frac{k}{m}v$. [3]
- (b) Given that the ball bearing is released at the surface of the liquid, show that $v = \frac{mg}{k} \left(1 - e^{-\frac{k}{m}t} \right)$. [9]
- (c) For a certain ball bearing, $m/k = 0.5$, find the velocity of this ball bearing when $t = 2$. [3]

Section (b): Statistics

8

- (a) When can the Poisson distribution be used as an approximation to Binomial distribution? [1]
- (b) It was found that 2.5% of tyres produced by a tyre company are defective. Calculate the probability that in a random sample of 100 tyres, there are
- (i) exactly 2 defective tyres, [2]
- (ii) at least one defective tyre. [3]

9

If $X \sim N(20; 3^2)$ and $Y \sim (32; 4^2)$, find

- (a) $P(X + Y > 54)$, [3]
- (b) $P(X > Y)$, [3]
- (c) $P(3Y - 2X > 60)$. [3]

- 10 (a) A soccer enthusiast records the number of goals, scored by his team. A random sample of 10 games played by the team gave the following goals:
1; 0; 3; 2; 5; 0; 2; 1; 4; 2
- (i) Calculate the unbiased estimate of the mean number of goals scored by his team.
- (ii) Prove that the unbiased estimate, σ^2 , of the variance is 2.667 correct to 3 decimal places. [3]
- (b) The random variable, X , has a normal distribution with mean $\mu = 50.7$ and standard deviation, σ . Given that $P(X < 60) = 0.79$, calculate the value of σ correct to 3 significant figures. [3]
- 11 (a) (i) Explain briefly what you understand by the term *random sample*.
- (ii) State any method of obtaining such a sample. [2]
- (b) 60 students from Isu High were invited to a party hosted by the local MP. To facilitate this, the Head divided the students into 6 groups, i.e. F1, F2, F3, F4, F5 and F6. He then chose 10 students at random from each form.
- (i) State the sampling method used by the Head.
- (ii) State an advantage of using this method. [2]
- (c) The time Form 6 students spend each day in the library is known to be normally distributed. A random sample of 60 students was taken and the time they spend in the library was found to have a mean time of 10 hours and standard deviation of 2 hours.
- Find, at 95% confidence interval, the mean time they spend in the library. [3]
- 12 The mass of a standard loaf of bread is normally distributed with mean 700 g. Customers complained that a loaf being sold at V.G. store is underweight. The standards control took a random sample of 64 loaves from the store, which gave a mean weight of 680 g and standard deviation 64 g.
- Show whether this provides evidence at 1% level of significance of a reduced mean mass. [6]

- 13 A survey was done to find how many smokers and non-smokers suffer from T.B. and the results of the survey are set out in the contingency table below.

	Suffering from T.B.	T.B. free	Total
Smokers	100	80	180
Non-smokers	120	200	320
Total	220	280	500

Test at 5% level of significance if there is an association between smokers and T.B.

[12]

- 14 A sociologist is in charge of 10 children and calculates the cost of keeping each child per week. The following results were obtained.

Age, x	1	2	3	4	2	5	7	8	5	2
Weekly cost	10	7	5	5	6	4	3	3	5	8

- (a) Draw a scatter diagram to show the above information. [3]
- (b) Find the regression line Y on X , hence estimate the cost of keeping a 6 year old child for a week. [7]
- (c) Calculate the product moment correlation coefficient and comment about the correlation. [4]

